FarmVerse – Precision Agriculture Management System

1. Project Overview

FarmVerse is a smart farming platform that helps farmers monitor their farms using IoT sensors, Artificial Intelligence (AI), weather forecasting, and cloud computing. The system collects real-time data from the farm, analyzes it, and provides intelligent recommendations for irrigation, fertilizer application, crop disease detection, and yield prediction. Instead of relying on traditional farming methods, FarmVerse enables farmers to make data-driven decisions, increasing productivity while reducing water usage, fertilizer waste, and crop losses.

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2. Problem Statement

Agriculture plays a vital role in food production, but farmers face several challenges such as irregular rainfall, water scarcity, poor soil monitoring, excessive fertilizer usage, delayed disease detection, and lack of real-time information. These issues reduce crop yield, increase farming costs, and lead to inefficient use of resources.

Most farming decisions are still based on manual observation and experience rather than real-time data. Therefore, there is a need for a smart precision agriculture system that continuously monitors field conditions, predicts potential problems, and provides timely recommendations to farmers.

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3. Proposed Solution

FarmVerse integrates IoT sensors, cloud computing, AI, and weather forecasting into one platform.

The system:

- Monitors soil moisture, temperature, humidity, and pH in real time.
- Collects weather forecasts using weather APIs.
- Uses AI to detect crop diseases from plant images.
- Recommends irrigation schedules.
- Suggests fertilizers based on soil conditions.
- Predicts crop yield.
- Sends notifications and alerts to farmers.
- Displays reports through a web or mobile application.

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4. Objectives

- Improve crop productivity.
- Reduce water wastage.
- Detect crop diseases early.
- Optimize fertilizer usage.

- Reduce farming costs.
- Support sustainable agriculture.
- Help farmers make better decisions using real-time data.

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5. Target Users

- Farmers
- Agricultural Officers
- Agronomists
- Farm Owners
- Agricultural Organizations

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6. Functional Requirements

User Module

- Registration
- Login
- Profile Management

Farm Module

- Add Farm
- Add Crop
- Farm Location
- Plot Management

IoT Module

- Soil Moisture Monitoring
- Temperature Monitoring
- Humidity Monitoring
- Soil pH Monitoring

Weather Module

- Live Weather
- Rain Forecast
- Temperature Forecast

AI Module

- Crop Disease Detection

- Yield Prediction
- Fertilizer Recommendation
- Irrigation Recommendation

Dashboard

- Sensor Values
- Crop Status
- Farm Analytics
- Reports

Notification Module

- Irrigation Alerts
- Rain Alerts
- Disease Alerts
- Harvest Alerts

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7. Non-Functional Requirements

- Secure Login
- User-friendly Interface
- Fast Response Time
- Real-time Monitoring
- Cloud Storage
- High Availability
- Scalability
- Reliable Data Backup

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8. Hardware Requirements

- ESP32
- Soil Moisture Sensor
- DHT11/DHT22 Sensor
- Soil pH Sensor
- Relay Module
- Water Pump
- Wi-Fi Module

9. Software Requirements

Frontend

- React.js
- Flutter (Optional)

Backend

- Node.js (Express)

Database

- MongoDB

AI

- Python
- TensorFlow
- OpenCV

Cloud

- Firebase
- AWS

APIs

- OpenWeather API

10. System Architecture

Farmer

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Web / Mobile Application

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(HTTP/HTTPS)

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Backend Server (API)

(Node.js / Spring Boot)

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↓	↓	↓	↓
Authentication	Farm Mgmt	IoT Service	Weather API

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AI / ML Engine

(Disease Detection • Yield Prediction • Recommendations)

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Database

(MongoDB / MySQL)

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Notification Service

(SMS • Email • Push Notifications)

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Farmer

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11. Backend Architecture

Client

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```
Express REST API
   |
   ├── Authentication Module
   ├── Farm Module
   ├── Sensor Module
   ├── Weather Module
   ├── AI Module
   ├── Recommendation Module
   ├── Notification Module
   └── Report Module
          ↓
      MongoDB
```

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12. Database Design

Users

- userId
- name
- email
- password
- role

Farms

- farmId
- userId
- location
- area

Crops

- cropId
- farmId
- cropName
- plantingDate

Sensors

- sensorId
- moisture
- temperature
- humidity

- pH
- timestamp

Weather

- weatherId
- rainfall
- temperature
- humidity

Disease Reports

- diseaseId
- cropName
- diseaseName
- treatment

Recommendations

- recommendationId
- irrigation
- fertilizer
- pesticide

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13. Workflow

1. Farmer logs into the application.
2. Registers farm details.
3. IoT sensors send live environmental data.
4. Backend stores the data.
5. Weather API provides forecast information.
6. AI analyzes sensor data and crop images.
7. The recommendation engine generates suggestions.
8. Notifications are sent to the farmer.
9. Reports and analytics are displayed on the dashboard.

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14. Expected Output

- Real-time soil monitoring
- Smart irrigation recommendations

- Fertilizer recommendations
- Disease detection
- Weather alerts
- Crop yield prediction
- Water usage reports
- Farm performance analytics

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15. Future Enhancements

- Drone-based crop monitoring
- Satellite image analysis
- Voice assistant in local languages
- Blockchain-based crop traceability
- Marketplace for farmers
- Smart irrigation automation
- AI chatbot for farming support

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16. Benefits

- Saves water through precision irrigation.
- Detects diseases early using AI.
- Optimizes fertilizer usage.
- Increases crop yield and quality.
- Reduces farming costs.
- Provides real-time monitoring and alerts.
- Supports sustainable agriculture.
- Enables data-driven decision making.